

EXPERIMENT-7

IMPLEMENTATION OF 2DES

AIM:

To implement Double DES (2DES) encryption and decryption using two different keys.

ALGORITHM

1. Read the plaintext from the user.
2. Read two keys, K1 and K2.
3. Encrypt the plaintext using K1.
4. Encrypt the result again using K2.
5. Display the encrypted text in hexadecimal format.
6. Decrypt the ciphertext using K2.
7. Decrypt the result again using K1.
8. Display the original plaintext.

Encryption:

$$C = E(K2, E(K1, P))$$

Decryption:

$$P = D(K1, D(K2, C))$$

CODE:

```
main.c
1 #include <stdio.h>
2 #include <string.h>
3 int main()
4 {
5     char text[100], k1[20], k2[20];
6     int i, len, n1, n2;
7     printf("Enter message: ");
8     scanf("%s", text);
9     printf("Enter key 1: ");
10    scanf("%s", k1);
11    printf("Enter key 2: ");
12    scanf("%s", k2);
13    len = strlen(text);
14    n1 = strlen(k1);
15    n2 = strlen(k2);
16    /* First encryption */
17    for(i = 0; i < len; i++)
18        text[i] ^= k1[i % n1];
19    /* Second encryption */
20    for(i = 0; i < len; i++)
21        text[i] ^= k2[i % n2];
22    printf("Encrypted Text: ");
23    for(i = 0; i < len; i++)
24        printf("%02X ", (unsigned char)text[i]);
25    /* Second decryption */
26    for(i = 0; i < len; i++)
27        text[i] ^= k2[i % n2];
28    /* First decryption */
```

OUTPUT:

Output

```
Enter message: hello  
Enter key 1: key  
Enter key 2: abc  
Encrypted Text: 62 62 76 66 68  
Decrypted Text: hello
```

```
=== Code Execution Successful ===
```

RESULT:

Thus, the Double DES (2DES) encryption and decryption process was successfully demonstrated using two keys.